

Aniket Kumar

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SUMMARY

Software Engineer with 4+ years of experience building scalable, high-availability distributed systems using C++, Python, Rust and Golang with a strong focus on performance, reliability, and system design.

EDUCATION

University at Buffalo| MS in Computer Science

- Coursework: Database Management, Computer Architecture, Operating System, Data Intensive, Deep Learning

Vellore Institute of Technology| B.Tech in Electrical Engineering

- Coursework: Modern Network Concepts, Data Structures and Algorithms, Object Oriented Programming

SKILLS

Programming Languages: C++, C, Python, Golang, C#, TypeScript, Java, SQL, JavaScript, Rust

Machine Learning / AI: LLMs, Generative AI, LangChain, LangGraph, Semantic Kernel, AutoGen, RAG, Transformers, MCP

Frameworks: Spring Boot, FastAPI, Django, gRPC, GraphQL, Streamlit, Node JS, Flutter, ReactJS, NextJS, TensorFlow, PyTorch

Distributed Systems: Microservices, AWS, Azure, GCP, Docker, Kubernetes(AKS/EKS), Helm, Prometheus, Grafana, Jenkins

Data Systems: Kafka, Amazon SQS, PostgreSQL, MongoDB, Redis, Vector Databases (ChromaDB, Pinecone), Spark, Cassandra

WORK EXPERIENCE

Software Engineer Intern | VelocityRoad.ai | New York, NY

Jul 2025 - Aug 2025

- **AskDB - NLP Powered Database SQL Agent: (LangChain, LangGraph, Amazon RDS, FastAPI, RAG, Agentic workflow, Python)** Built an AI agent that analyzes database tables and schemas, retrieves relevant data, and generates reports from user queries using LLMs and training examples, enabling 50+ non-technical users to self-serve data and cut report creation time by 75%.

Senior Software Engineer | Wipro Technology

Dec 2023 - July 2024

- **Infrastructure Scaling & Traffic Surge Handling: (Go, Jenkins, AWS Secrets Manager, Cron)** Engineered a cache-aware auto-scaling system leveraging cache expiry to dynamically scale containers during traffic spikes, cutting API timeouts by 85% and cold-start latency 40% on Amazon ECS with Grafana-based monitoring.
- **Performance Optimization: (Golang, Python, gRPC, SQL, Typescript, ReactJS, HTML, CSS, JavaScript)** Migrated REST API from Python to Go (gRPC), reducing processing time by 80%, averaging API calls under 40 ms via indexing, concurrent batch processing, caching, and wrote Java (Maven) integration tests to validate cross-service workflows.
- **API Optimization: (Python, GraphQL, REST API)** Built a GraphQL microservice to unify cart items, promotions, inventory, and multi-fulfillment options like shipping, delivery, minimizing frontend API calls by 70% and improving cart page load time by 40%.
- **Reliability Engineering & Observability: (Alertmanager, ServiceNow, Grafana, Prometheus, Datadog)** Improved reliability with real-time alerts and reduced costs by 20% by cutting memory usage 60% through Alpine Docker images.
- **Time Series Forecasting: (Predictive Modelling, ML Ops Pipeline, Docker)** Built an end-to-end time-series forecasting pipeline on AWS leveraging ARIMA, XGBoost, and LSTM ensembles to predict price trends. Improved forecast accuracy by 18% over baseline and optimized procurement timing, saving \$2.1M annually.

Software Engineer | Wipro Technology

Aug 2020 - Dec 2023

- **Data Migration: [SAP HANA, SAP RAR]** Built end to end ETL pipelines to extract, transform, and load legacy financial data into SAP RAR on S4HANA, centralizing finance data and ensuring compliance with modern accounting standards.
- **Event Driven Distributed System Data Pipeline: [Airflow, Azure Data Factory, Data Lake, Azure Event Hubs]** Designed and built distributed Azure Data Factory pipelines to ingest unstructured data into a raw data lake, transform it into structured formats, processing 1TB+ quarterly and improving throughput by 45% and uptime by 60%.
- **Pipeline Automation: [Azure Synapse, Azure SQL]** Automated end-to-end ETL workflows, cutting manual effort by 100%, reducing processing time by ~50%, and delivering transformed data to downstream teams for further analysis.
- **Cloud Infrastructure and Containerization: (CI/CD, Helm, Terraform, GitHub Actions, Spinnaker)** Deployed apps (180+ containers) on Azure, automating auto-scaling and load balancing with Kubernetes. Handled 1M+ RPM with zero-downtime.

PROJECTS

TacoDB - Distributed Relational Database(C++) [[Link](#)]

- **[C, C++, Systems, Storage Engine, Linux, B+ Tree]** Designed and implemented a high-performance mini relational database engine with custom storage layouts, in-memory buffer management, B-Tree indexing, and optimized query processing.

Therapy.ai [[Link](#)][[Link](#)]

- **[FastAPI, Ollama, Lora, Hugging Face, Typescript, ReactJS, GenAI, , ReactJS, HTML, CSS, JavaScript]** Spearheaded the development of an AI-powered mental health companion prototype with real-time voice and video interaction, fine-tuning Mistral-7B using LoRA for empathetic responses and achieving ~85% positive user feedback.

Autonomous Driving using Reinforcement Learning [[Link](#)]

- **[DQN, DoubleDQN, TRPO, PPO, A2C, A3C algorithms, MetaDrive]** Implemented a training pipeline with LiDAR, camera, and vector data, boosting navigation accuracy by 25% through reward structure and experience replay optimization.

ACHIEVEMENTS

- **Publication** : Automatic Vegetable Differentiator in 2017 IEEE ((<https://ieeexplore.ieee.org/document/8391985>)).